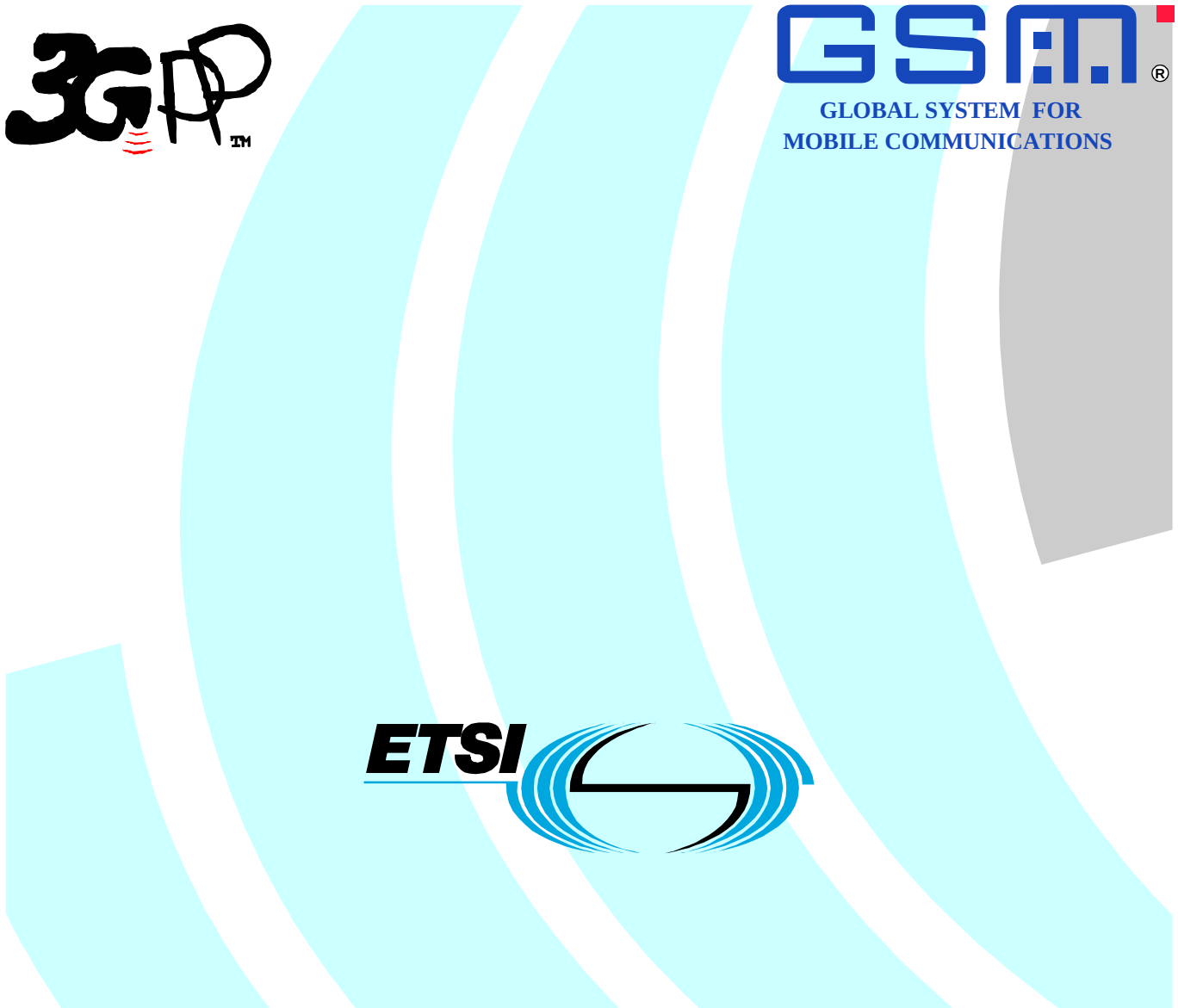
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Foreword

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1 Scope

The modulator receives the bits from the encryption unit, see 3GPP TS 45.001, and produces an RF signal. The filtering of the Radio Frequency (RF) signal necessary to obtain the spectral purity is not defined, neither are the tolerances associated with the theoretical filter requirements specified. These are contained in 3GPP TS 45.005.

1.1 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 45.001: "Physical Layer on the Radio Path (General Description)".

[3] 3GPP TS 45.002: "Multiplexing and multiple access on the radio path".

[4] 3GPP TS 45.005: "Radio transmission and reception".

1.2 Abbreviations

Abbreviations used in this specification are listed in 3GPP TR 21.905.

2 Modulation format for GMSK

2.1 Modulating symbol rate

The modulating symbol rate is 1/T = 1 625/6 ksymb/s (i.e. approximately 270.833 ksymb/s), which corresponds to 1 625/6 kbit/s (i.e. 270.833 kbit/s). T is the symbol period.

2.2 Start and stop of the burst

Before the first bit of the bursts as defined in 3GPP TS 45.002 enters the modulator, the modulator has an internal state as if a modulating bit stream consisting of consecutive ones ($d_i = 1$) had entered the differential encoder. Also after the last bit of the time slot, the modulator has an internal state as if a modulating bit stream consisting of consecutive ones ($d_i = 1$) had continued to enter the differential encoder. These bits are called dummy bits and define the start and the stop of the active and the useful part of the burst as illustrated in figure 1. Nothing is specified about the actual phase of the modulator output signal outside the useful part of the burst.

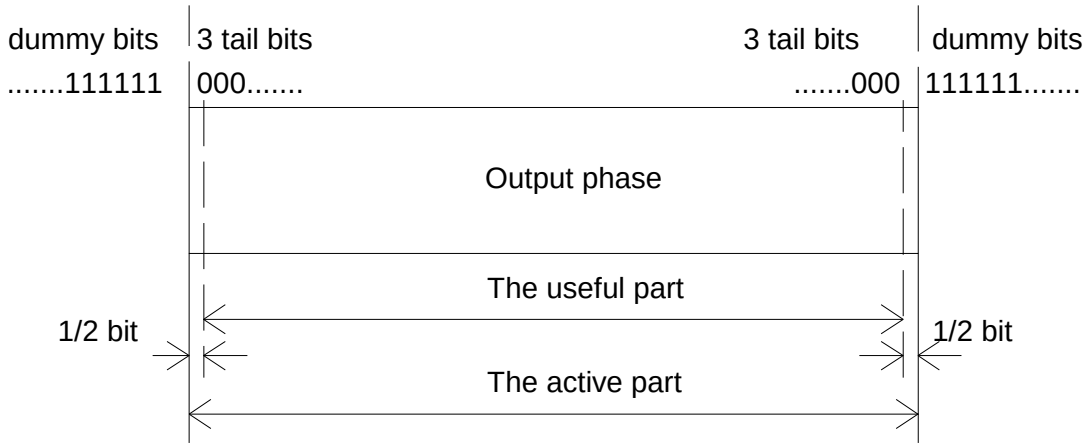


Figure 1: Relation between active part of burst, tail bits and dummy bits. For the normal burst the useful part lasts for 147 modulating bits

2.3 Differential encoding

Each data value $d_i = [0,1]$ is differentially encoded. The output of the differential encoder is:

$$\hat{d}_i = d_i \oplus d_{i-1} \qquad (d_i \in \{0,1\})$$

where $\oplus$ denotes modulo 2 addition.

The modulating data value α_i input to the modulator is:

$$\alpha_i = 1 - 2\hat{d}_i \qquad (\alpha_i \in \{-1,+1\})$$

2.4 Filtering

The modulating data values α_i as represented by Dirac pulses excite a linear filter with impulse response defined by:

$$g(t) = h(t) * rect\left(\frac{t}{T}\right)$$

where the function *rect(x)* is defined by:

$$rect\left(\frac{t}{T}\right) = \frac{1}{T} \qquad for\ |t| < \frac{T}{2}$$

$$rect\left(\frac{t}{T}\right) = 0 \qquad otherwise$$

and * means convolution. *h(t)* is defined by:

$$h(t) = \frac{\exp\left(\frac{-t^2}{2\delta^2 T^2}\right)}{\sqrt{(2\pi)}\ .\ \delta T}$$

where $\delta = \frac{\sqrt{\ln(2)}}{2\pi BT}$ *and* $BT = 0.3$

where B is the 3 dB bandwidth of the filter with impulse response $h(t)$. This theoretical filter is associated with tolerances defined in 3GPP TS 45.005.

2.5 Output phase

The phase of the modulated signal is:

$$\varphi(t') = \sum_i \alpha_i \pi h \int_{-\infty}^{t'-iT} g(u)du$$

where the modulating index h is 1/2 (maximum phase change in radians is π/2 per data interval).

The time reference $t' = 0$ is the start of the active part of the burst as shown in figure 1. This is also the start of the bit period of bit number 0 (the first tail bit) as defined in 3GPP TS 45.002.

2.6 Modulation

The modulated RF carrier, except for start and stop of the TDMA burst may therefore be expressed as:

$$x(t') = \sqrt{\frac{2E_c}{T}} \cdot \cos(2\pi f_0 t' + \varphi(t') + \varphi_0)$$

where E_c is the energy per modulating bit, f_0 is the centre frequency and φ_0 is a random phase and is constant during one burst.

3 Modulation format for 8PSK

3.1 Modulating symbol rate

The modulating symbol rate is 1/T = 1 625/6 ksymb/s (i.e. approximately 270.833 ksymb/s), which corresponds to 3*1 625/6 kbit/s (i.e. 812.5 kbit/s). T is the symbol period.

3.2 Symbol mapping

The modulating bits are Gray mapped in groups of three to 8PSK symbols by the rule

$$s_i = e^{j2\pi l/8}$$

where l is given by table 1.

Table 1: Mapping between modulating bits and the 8PSK symbol parameter *l*.

Modulating bits d_{3i}, d_{3i+1}, d_{3i+2}	Symbol parameter l
(1,1,1)	0
(0,1,1)	1
(0,1,0)	2
(0,0,0)	3
(0,0,1)	4
(1,0,1)	5
(1,0,0)	6
(1,1,0)	7

This is illustrated in figure 2.

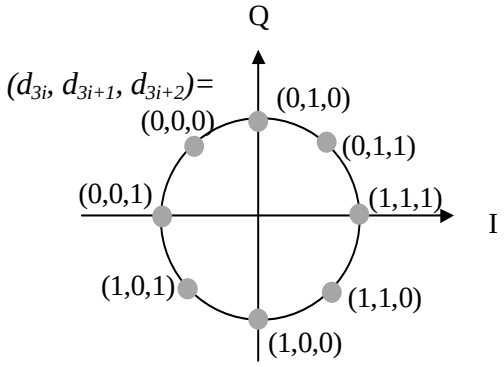


Figure 2: Symbol mapping of modulating bits into 8PSK symbols.

3.3 Start and stop of the burst

Before the first bit of the bursts as defined in 3GPP TS 45.002 enters the modulator, the state of the modulator is undefined. Also after the last bit of the burst, the state of the modulator is undefined. The tail bits (see 3GPP TS 45.002) define the start and the stop of the active and the useful part of the burst as illustrated in figure 3. Nothing is specified about the actual phase of the modulator output signal outside the useful part of the burst.

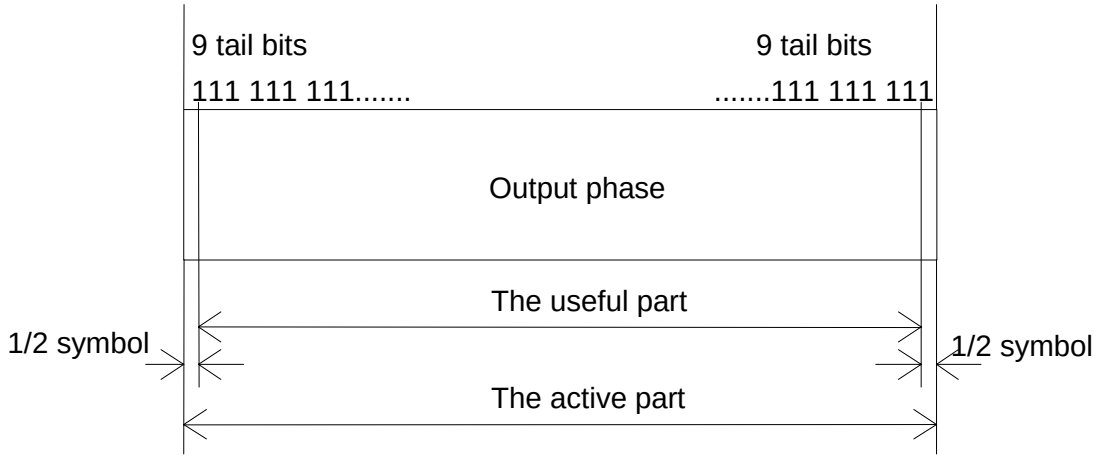


Figure 3: Relation between active part of burst and tail bits. For the normal burst the useful part lasts for 147 modulating symbols

3.4 Symbol rotation

The 8PSK symbols are continuously rotated with 3π/8 radians per symbol before pulse shaping. The rotated symbols are defined as

$$\hat{s}_i = s_i \cdot e^{ji3\pi/8}$$

3.5 Pulse shaping

The modulating 8PSK symbols $\hat{s}_i$ as represented by Dirac pulses excite a linear pulse shaping filter. This filter is a linearised GMSK pulse, i.e. the main component in a Laurant decomposition of the GMSK modulation. The impulse response is defined by:

$$c_0(t) = \begin{cases} \prod_{i=0}^{3} S(t+iT), \ for\ 0 \le t \le 5T \\ 0, \ else \end{cases}$$

where

$$S(t) = \begin{cases} \sin(\pi \int_0^t g(t')dt'), \ for\ 0 \le t \le 4T \\ \sin(\frac{\pi}{2} - \pi \int_0^{t-4T} g(t')dt'), \ for\ 4T < t \le 8T \\ 0, \ else \end{cases}$$

$$g(t) = \frac{1}{2T}\left(Q(2\pi \cdot 0.3\frac{t-5T/2}{T\sqrt{\log_e(2)}}) - Q(2\pi \cdot 0.3\frac{t-3T/2}{T\sqrt{\log_e(2)}}) \right)$$

and

$$Q(t) = \frac{1}{\sqrt{2\pi}} \int_t^{\infty} e^{-\frac{\tau^2}{2}} d\tau .$$

The base band signal is

$$y(t') = \sum_i \hat{s}_i \cdot c_0(t' - iT + 2T)$$

The time reference $t' = 0$ is the start of the active part of the burst as shown in figure 3. This is also the start of the symbol period of symbol number 0 (containing the first tail bit) as defined in 3GPP TS 45.002.

3.6 Modulation

The modulated RF carrier during the useful part of the burst is therefore:

$$x(t') = \sqrt{\frac{2E_s}{T}} \mathrm{Re}\left[y(t') \cdot e^{j(2\pi f_0 t' + \varphi_0)} \right]$$

where E_S is the energy per modulating symbol, f_0 is the centre frequency and φ_0 is a random phase and is constant during one burst.

Annex A (informative): Change history

SMG	SPEC	VERS	NEW_VE	PHA	SUBJECT
S27	05.04	5.0.1	6.0.0	R97	Conversion to Release 97 EN
S28	05.04	6.0.0	8.0.0	R99	Introduction of 8PSK for EDGE
S30b	05.04	8.0.0	8.1.0	R99	Correction of mistake for range alpha-sub-i in Clause 2.3
		8.1.0	8.1.1		Figure 3 replaced (as it was corrupted)
G03	05.04	8.1.1	8.2.0	R99	Correction of symbol period notation
G04	45.004	8.2.0	4.0.0	Rel-4	New version for Release 4
G05	45.004	4.0.0	4.1.0	Rel-4	Correction of Timing Alignment for GMSK and 8-PSK Signals

Change history							
Date	TSG #	TSG Doc.	CR	Rev	Subject/Comment	Old	New
2001-11	07	GP-012359	002		Correction of tail bits for 8PSK normal burst	4.1.0	4.2.0
2001-11	07	GP-012372	004		Correction of references to relevant 3GPP TSs	4.1.0	4.2.0
2001-11	07	GP-012360	003		Correction of tail bits for 8PSK normal burst	4.2.0	5.0.0
2002-06	10	GP-021436	005		Corrections and clean up	5.0.0	5.1.0
2003-09					Correction of wrong Release number in the front page	5.1.0	5.1.1

History

Document history		
V5.0.0	November 2001	Publication
V5.1.0	June 2002	Publication (Withdrawn)
V5.1.1	September 2003	Publication